USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 24 LAB - Null Steering
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.9: I can calculate null-steering weights and implement pattern nulls on the ADALM-PHASER.

This packet is the L24 Lab, in three parts, and you turn it in once, complete, before Lesson 25. **Part 1, Predict**, is yours to do before the lab period: what each Part 2 table should read on your kit, and the gains and phases you type in for Procedure A. **Part 2, Measure**, is your own kit's readings, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict Tables A to D.** Fill in each table below with what the matching Part 2 table should read on your kit, at the HB100 frequency your kit reported with **Find HB100** in an earlier lab. If you did not record one, use the nominal 10.525 GHz and say so. Lesson 24's Pre-Lab frame names the frame that works each method, at 10.0 GHz.

   The array has $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch, with $+\theta$ on the E8 side; use $c = 3.00 \times 10^8$ m/s. Each ADAR1000 sums four patches, so the two digital channels are E1 to E4 and E5 to E8. Its phase shifters step in 2.8125° and its gains in 1% of full scale, and the Element Gains sliders are linear in voltage. Procedure A holds the beam at broadside and nulls $\theta_1 = +22.5^\circ$. In Procedure C the HB100 is the target, at boresight, and a simulated interferer arrives 10 dB stronger, first at $+20^\circ$ and then at $+30^\circ$. Report levels as Part 2 does: Tables A and B relative to the frozen uniform peak, and Peak Gain in Table C as its change from Manual's.

Frequency								
Steps 7, 8: element	1	2	3	4	5	6	7	8
Gain (%)								
Phase (°)								

Table A	**Uniform SL nearest $+22.5^\circ$ (dBc)**	**Null-steered peak (dB)**	**Notch bottom (dB)**	**Notch angle**

Table B	**Null depth (dB)**	**Null angle**	**Left peak angle**	**Right peak angle**

Table C, $+20^\circ$	**Manual, Int. vs Target**	**MVDR, Peak Gain**	**MVDR null angle**	**Conventional at null**

Table C, $+30^\circ$	**MVDR** 0.001**, Peak Gain**	**MVDR** 0.1**, Peak Gain**	

Table D	**Delta null depth (dBc)**	**Delta null angle**	**Error zero crossing**	**Error sign below, above**

Table D, stand	-20°	-10°	$+10^\circ$	$+20^\circ$
Tracked angle				
How far				

Show your work for one row of each table. For the gains and phases, your script is enough.

Part 2: Measure (your kit, in class)

Follow the procedure on the L24 Lab page, and record each reading as you take it.

Freq = | Rx Gain = | Distance =

2. **Table A: the static notch** (Procedure A, Steps 6–10). Freeze the uniform trace first. If its two first sidelobes differ by more than about 2 dB , recalibrate before you go on. Read peaks with **Peak Gain** and angles with **Angle** set to **Peak**.

Uniform SL nearest $+22.5^\circ$ **(dBc)**	**Null-steered peak (dB)**	**Notch bottom (dB)**	**Notch angle**

3. **Table B: the difference null** (Procedure B, Steps 11–15). Beam 1 Phase at 180°, Mode Manual, uniform taper.

Null depth (dB)	Null angle	Left peak angle	Right peak angle

4. **Table C: MVDR against an interferer** (Procedure C, Steps 16–22). Simulated interferer at $+20^\circ$, Power 0 dBFS, about 10 dB above the target; target your HB100 at boresight; beam held at 0° in Signal vs Time; MVDR with $K = 1024$ snapshots. Each entry is the median of ten readings.

Mode	Interferer vs Target	Peak Gain	MVDR Suppression
Manual			
MVDR, load 0.1			

Rectangular, held	Null angle	Null depth	Conventional at null
MVDR, load 0.1			

Interferer at $+30^\circ$	Manual	MVDR, load 0.001	MVDR, load 0.1
Peak Gain			

5. **Table D: monopulse** (Procedure D, Steps 23–27). The Tracking preset, Manual beamformer, both digital beams at 0°, uniform taper.

Delta null depth (dBc)	Delta null angle	Error zero crossing	Error sign below, above

Stand's scale	-20°	-10°	$+10^\circ$	$+20^\circ$
Tracked angle				
How far it followed				

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

6. **Three nulls.** Compare your notch (Table A), your difference null (Table B) and your delta null (Table D) with each other and with your predictions. Explain what set the depth and the angle of each one on your kit, and point to the readings that support each explanation.

7. **MVDR at two angles.** Compare your Manual and MVDR readings with the interferer at $+20^\circ$ and at $+30^\circ$ (Table C), and with your predictions. Explain what MVDR did at each angle. The interferer is simulated: say which of your conclusions would hold for a real interferer in this room, and what Table C doesn't explain about one.

8. **The tracker.** Compare your Table D readings with your predictions: the beam sweep of Steps 25 and 26, and the tracked angles of Step 27 against the stand's scale. How closely did the loop point the beam at the HB100, and what set that accuracy? From what you saw as you moved the HB100, say what limits how fast a source this loop can follow.

Documentation: _____